

```
from weasyprint import HTML
```

```
# Defining the HTML content for the Requirements Document in English
```

```
html_content = """
```

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <style>
```

```
    @page {
```

```
      size: A4;
```

```
      margin: 20mm;
```

```
      background-color: #fcfcfc;
```

```
    }
```

```
    body {
```

```
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
```

```
      color: #333;
```

```
      line-height: 1.6;
```

```
      margin: 0;
```

```
      padding: 0;
```

```
    }
```

```
    .header-banner {
```

```
      background-color: #2c3e50;
```

```
      color: white;
```

```
      padding: 30px;
```

```
      text-align: center;
```

```
      margin-bottom: 30px;
```

```
    }
```

```
    h1 {
```

```
      margin: 0;
```

```
      font-size: 22pt;
```

```
      text-transform: uppercase;
```

```
      letter-spacing: 1px;
```

```
    }
```

```
    .project-meta {
```

```
      margin-top: 10px;
```

```
      font-size: 11pt;
```

```
      opacity: 0.9;
```

```
    }
```

```
    .section {
```

```
      margin-bottom: 25px;
```

```
      page-break-inside: avoid;
```

```
    }
```

```
    h2 {
```

```
      color: #2980b9;
```

```
      font-size: 16pt;
```

```
      border-left: 5px solid #2980b9;
```

```
      padding-left: 10px;
```

```

        margin-top: 20px;
        margin-bottom: 15px;
    }
    .sprint-objective {
        font-style: italic;
        background-color: #ebf5fb;
        padding: 10px 15px;
        border-radius: 4px;
        margin-bottom: 15px;
        font-size: 10.5pt;
    }
    .us-card {
        background-color: white;
        border: 1px solid #e1e4e8;
        padding: 12px;
        margin-bottom: 10px;
        border-radius: 4px;
    }
    .us-title {
        font-weight: bold;
        color: #2c3e50;
        display: block;
        margin-bottom: 5px;
    }
    .ac-list {
        margin-top: 8px;
        margin-bottom: 0;
        padding-left: 20px;
        font-size: 10pt;
        color: #555;
    }
    .ac-item {
        margin-bottom: 4px;
    }
    footer {
        position: fixed;
        bottom: 0;
        width: 100%;
        text-align: center;
        font-size: 9pt;
        color: #95a5a6;
        padding-bottom: 10px;
    }
</style>
</head>
<body>
    <div class="header-banner">
        <h1>Requirements Document</h1>

```

```
<div class="project-meta">
  <strong>Project:</strong> MedManager (Local C++ Vaccination System) |
<strong>Version:</strong> 1.0
</div>

<div class="section">
  <p><strong>Scope:</strong> This document specifies the software requirements for a
stand-alone management system for a single vaccination center. It focuses on physical
vaccine stock control, appointment scheduling, and the end-to-end vaccination process.</p>
</div>

<div class="section">
  <h2>Sprint 1: Project Foundation & Modeling</h2>
  <div class="sprint-objective">
    Objective: Setup the development environment and establish core software
engineering artifacts tailored to the local system.
  </div>

  <div class="us-card">
    <span class="us-title">US01 - Artifact Maintenance</span>
    The team must update all existing artifacts during each sprint to accurately reflect
changes in business needs.
  </div>
  <div class="us-card">
    <span class="us-title">US02 - Traceability</span>
    For each user story, the team must elaborate artifacts covering requirements,
analysis, design, testing, and implementation.
  </div>
  <div class="us-card">
    <span class="us-title">US03 - Repository Setup</span>
    The team must setup its own working environment (e.g., Git/GitHub repository).
  </div>
  <div class="us-card">
    <span class="us-title">US04 - Glossary</span>
    Construct a technical glossary for the current project.
  </div>
  <div class="us-card">
    <span class="us-title">US05 - Requirements Investigation</span>
    Capture functional and non-functional requirements using a Use Case Diagram.
  </div>
  <div class="us-card">
    <span class="us-title">US06 - Domain Model</span>
    Elaborate a Domain Model that reflects the specific reality of the local vaccination
center.
  </div>
</div>
```

```
<div class="section">
  <h2>Sprint 2: Internal Management (Admin & Inventory)</h2>
  <div class="sprint-objective">
    Objective: Implement features to populate the system with vaccine types, physical
    stock (doses), and employee records.
  </div>

  <div class="us-card">
    <span class="us-title">US10 - Specify Vaccine Type</span>
    As a <strong>CenterAdministrator</strong>, I want to specify a new vaccine type in
    the system.
  </div>
  <div class="us-card">
    <span class="us-title">US11 - Register Physical Vaccine Stock</span>
    As a <strong>CenterAdministrator</strong>, I want to register a physical vaccine
    dose in the inventory.
    <ul class="ac-list">
      <li class="ac-item"><strong>AC11-1:</strong> Registration must include lot
      number, brand, and expiration date, linked to an existing vaccine type.</li>
    </ul>
  </div>
  <div class="us-card">
    <span class="us-title">US12 - Consult Inventory</span>
    As a <strong>CenterAdministrator</strong>, I want to list all physical vaccine stock
    available.
    <ul class="ac-list">
      <li class="ac-item"><strong>AC12-1:</strong> Vaccines must be grouped by type
      and listed alphabetically by brand name.</li>
    </ul>
  </div>
  <div class="us-card">
    <span class="us-title">US14 - Register Employee</span>
    As a <strong>CenterAdministrator</strong>, I want to register a new employee.
    <ul class="ac-list">
      <li class="ac-item"><strong>AC14-1:</strong> Each employee must be assigned
      a single role (Nurse, Receptionist, or Administrator).</li>
      <li class="ac-item"><strong>AC14-2:</strong> Name, Phone, and Email are
      mandatory fields.</li>
      <li class="ac-item"><strong>AC14-3:</strong> Phone and Citizen Card Number
      must comply with the Portuguese format.</li>
      <li class="ac-item"><strong>AC14-4:</strong> Phone and Email must be unique
      across all registered employees.</li>
    </ul>
  </div>
</div>

<div class="section">
  <h2>Sprint 3: User Interaction & Scheduling</h2>
```

```
<div class="sprint-objective">
  Objective: Develop the reception module for user enrollment and appointment
  creation.
</div>

<div class="us-card">
  <span class="us-title">US20 - Register SNS User</span>
  As a <strong>Receptionist</strong>, I want to register a new SNS user.
  <ul class="ac-list">
    <li class="ac-item"><strong>AC20-1:</strong> Phone and Citizen Card Number
    must comply with the Portuguese format.</li>
    <li class="ac-item"><strong>AC20-2:</strong> Sex attribute is optional; all other
    fields are mandatory.</li>
    <li class="ac-item"><strong>AC20-3:</strong> Phone, Email, CC Number, and
    SNS Number must be unique.</li>
  </ul>
</div>

<div class="us-card">
  <span class="us-title">US21 - Assisted Scheduling</span>
  As a <strong>Receptionist</strong>, I want to schedule a vaccine administration for
  an SNS user (in-person or phone).
</div>

<div class="us-card">
  <span class="us-title">US30 - Self-Scheduling</span>
  As an <strong>SNS User</strong>, I want to use the application to schedule my own
  vaccine for a specific date.
</div>
</div>

<div class="section">
  <h2>Sprint 4: The Vaccination Flow</h2>
  <div class="sprint-objective">
    Objective: Orchestrate the day-of-vaccination cycle, from arrival to administration and
    stock update.
  </div>

  <div class="us-card">
    <span class="us-title">US22 - Register Arrival</span>
    As a <strong>Receptionist</strong>, I want to register the arrival of an SNS user at
    the center.
    <ul class="ac-list">
      <li class="ac-item"><strong>AC22-1:</strong> Prevent duplicate entries for the
      same user on the same day.</li>
      <li class="ac-item"><strong>AC22-2:</strong> The user must be assigned to the
      "Waiting Room" status.</li>
    </ul>
  </div>
</div>
```

```

        <span class="us-title">US40 - Consult Waiting Room</span>
        As a <strong>Nurse</strong>, I want to consult the list of users in the waiting room.
        <ul class="ac-list">
            <li class="ac-item"><strong>AC40-1:</strong> Users must be listed on a
First-Come, First-Served basis.</li>
        </ul>
    </div>
    <div class="us-card">
        <span class="us-title">US41 - Record Administration</span>
        As a <strong>Nurse</strong>, I want to record the administration of a vaccine.
        <ul class="ac-list">
            <li class="ac-item"><strong>AC41-1:</strong> The nurse must select the specific
lot number from the inventory. The system must remove that dose from stock upon
confirmation.</li>
            <li class="ac-item"><strong>AC41-2:</strong> The user must be assigned to the
"Recovery Room" status.</li>
        </ul>
    </div>
</div>

<footer>
    MedManager Project - Requirements Specification v1.0
</footer>
</body>
</html>
"""

```

Generating the PDF

```
input_html_path = 'requirements.html'
```

```
output_pdf_path = 'Requirements_Document_MedManager.pdf'
```

```
with open(input_html_path, 'w', encoding='utf-8') as f:
```

```
    f.write(html_content)
```

```
HTML(filename=input_html_path).write_pdf(output_pdf_path)
```